

Cross-Asset Volatility and Equity Risk

Research proposal

Quantitative Techniques for Financial Economics
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Research question and motivation

Does past volatility in interest rates and oil prices improve forecasts of future S&P 500 volatility beyond the information in its own past volatility?

Interest rates

Volatile Treasury yields may reflect uncertainty about monetary policy and the discount rates used to value equities.

Oil prices

Volatile oil prices may reflect uncertainty about inflation and firms' energy costs.

Why it matters: Better forecasts of equity volatility could help investors monitor portfolio risk and make hedging decisions.

Do bond and energy signals add predictive information?

SUCCESSIVE MODELS

1. Equity baseline

Past realised equity volatility e.g. S&P 500

2. + Bond-market volatility

Baseline plus a measure of bond-market volatility

3. + Energy-market volatility

Adds a measure of energy-market volatility

HYPOTHESES

H1 Bond-market volatility contains information about subsequent equity volatility beyond the equity baseline.

H2 Energy-market volatility adds further predictive information.

Criterion: We will assess whether the additional variables improve forecasts on data held out from model estimation.

Daily market data provide a feasible starting point

ROLE	SERIES AND CONSTRUCTION
Equity outcome	S&P 500 daily returns; realised volatility over the following period
Equity baseline	Past realised S&P 500 volatility
Bond-market signal	US Treasury yields: realised volatility of daily yield changes, tested for several maturities (e.g. 2-, 5- and 10-year)
Energy-market signal	Brent crude oil prices: realised volatility of daily returns



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